

Will Kilgore

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Education

McGill University

Bachelor of Engineering in Mechanical Engineering - GPA: 3.84/4.0

Expected Dec 2027

Montreal, QC

Work Experience

Automated Tire, Inc.

May 2026 – Aug 2026

Mechanical Engineering Intern – Autonomous tire-changing robotic platform

Woburn, MA

- Designed, fabricated, and validated a custom life-cycle testing jig, sourcing components and supplier quotes to evaluate tire-rotation mechanism durability under repeated loading.
- Created technical drawings and assembly documentation in Onshape for 100+ component production assemblies, enabling technician-led builds without engineer oversight and reducing assembly time.
- Designed, fabricated, and iterated process-specific fixtures and tooling — selecting manufacturing processes based on cost, lead time, and production volume — to resolve assembly issues and reduce production time by 40%
- Simplified top-level assembly CAD by implementing a fastener-visibility toggle, reducing mate solving time from 3.5s to 1.2s (66% faster) across 4,000+ fastener instances, improving CAD responsiveness for design team
- Identified and documented design discrepancies during robotic system assembly, provided structured redline feedback, and processed ECOs that resolved recurring build issues across 2 design iterations
- Maintained three multi-material FDM printers including calibration, hardware servicing, and failure diagnosis, reducing print failures and achieving same-day prototype turnaround for engineering team

Parlee Composites

June 2025 – Present

Manufacturing and Engineering Intern – Carbon fiber bicycle design and manufacturing

Beverly, MA

- Conducted experimental vibration analysis using accelerometer data and analytical models to evaluate bicycle frame compliance, informing design decisions for a new frame platform
- Developed a graphical fit-analysis system overlaying generated rider fit coordinates with 200 existing fits, reducing required production part count by 25% while maintaining rider coverage.
- Implemented an Excel-based fit analysis tool generating 150+ stem stack/reach coordinate combinations across varying stem lengths, stem angles, and frame sizes to reduce dealer consultation time
- Introduced new prepreg composite layups that reduced void content and post-processing time, increasing small-part production throughput by up to 30%
- Managed quality control for overseas frame production using inspection checklists and defect tracking, reducing recurring supplier issues through improved defect identification and feedback loops

Projects

McGill Rocket Team

Sept 2023 – Present

Airframe Composites Project Co-Lead

- Co-led composites manufacturing for the aerostructures subteam, coordinating layups, tooling, and assembly integration across a 10-member subteam to deliver flight-ready structures for Launch Canada
- Designed and fabricated molds for a 42" composite nose cone as part of the aerostructures subteam, enabling the transition to a fully student-built 8" rocket platform
- Executed vacuum-assisted resin infusion (VARI) and prepreg layups to manufacture lightweight, high-strength structural body tubes for competition rockets

Skills

Technical Skills: Rapid Prototyping (3D Printing), Design for Manufacturing & Assembly (DFMA), GD&T, FEA, Vibrational Analysis, Experimental Design, Composite Manufacturing (Prepreg, VARI), Cable Routing & Harnessing

Software: CAD (SolidWorks, Onshape), Lifecycle Management (Arena), FEA (Abaqus), Python, Excel, AI-assisted workflows